

The Poisson Process

PIPE - MESIO
Departament de Matemàtiques, UPC

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1 The Poisson Process

A system is considered in which a population of users arrives independently at random times within a fixed interval $[0, T]$. The arrival time for each user is modelled as a random variable uniformly distributed over this interval. Consequently, the probability that a user arrives during a subinterval $I_{\Delta t}$ of length Δt is $p = \Delta t/T$. Let n denote the total number of users, and let $N_{\Delta t}$ represent the number of users arriving within $I_{\Delta t}$. Given the independence of arrivals, $N_{\Delta t}$ follows a binomial distribution with parameters n and p :

$$\mathbb{P}(N_{\Delta t} = k) = \binom{n}{k} \left(\frac{\Delta t}{T}\right)^k \left(1 - \frac{\Delta t}{T}\right)^{n-k}, \quad k = 0, 1, \dots, n.$$

For large values of n and T , the above probability will be well approximated by the Poisson distribution. Specifically, let n and T tend to infinity so that the number n/T of users arriving per unit of time tends to a constant value λ . Since

$$n \rightarrow \infty, \quad p = \frac{\Delta t}{T} \rightarrow 0, \quad np = \frac{n\Delta t}{T} \rightarrow \lambda\Delta t,$$

we can apply the Poisson theorem to find that, for any fixed $k \geq 0$:

$$\mathbb{P}(N_{\Delta t} = k) \rightarrow e^{-\lambda\Delta t} \frac{(\lambda\Delta t)^k}{k!}.$$

That is, in the limit, the random variable $N_{\Delta t}$ follows a Poisson law with parameter $\lambda\Delta t$ (convergence in distribution).

Here it is worth noting that if Δt is small, then

- (i) $\mathbb{P}(N_{\Delta t} = 1) = e^{-\lambda\Delta t} \lambda\Delta t \equiv \lambda\Delta t + o(\Delta t)$,
- (ii) $\mathbb{P}(N_{\Delta t} = 0) = e^{-\lambda\Delta t} \equiv 1 - \lambda\Delta t + o(\Delta t)$,
- (iii) $\mathbb{P}(N_{\Delta t} = k) = o(\Delta t)$, for $k \geq 2$.

1.1 Definition

If $I_{\Delta t}$ is the interval $(0, t]$, then $N_{\Delta t} \equiv N(t)$ counts the number of transitions (users who accessed the system) until the moment t , with the initial condition $N(0) = 0$. The stochastic process $N(t)$ obtained in this way is known as the *Poisson process of intensity* (or *rate*) λ . The process $N(t)$ is continuous in time and discrete in states, and its first-order probability function is given by:

$$\mathbb{P}(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \dots \quad (1)$$

Equation (1) illustrates how the probability function of $N(t)$ evolves over time. For each fixed $t > 0$, $N(t)$ behaves as a Poisson random variable with parameter λt . Figure 1 depicts a potential realization of this counting process, displaying the random points in time when transitions occur and the corresponding increases in the state of the process.

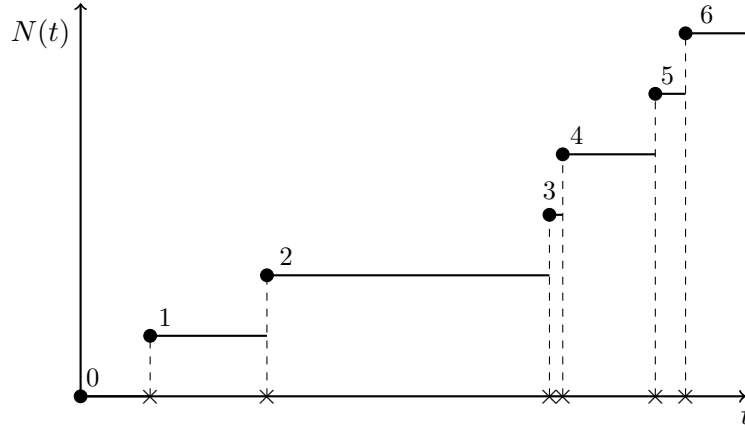


Figure 1: A realization of a Poisson process with intensity $\lambda = 0.4$ for $t \in [0, 15]$

For any times $t_1 < t_2$, the difference $N(t_2) - N(t_1)$ denotes the number of transitions occurring in the process during the interval $(t_1, t_2]$. As established previously, $N(t_2) - N(t_1)$ follows a Poisson distribution with parameter $\lambda(t_2 - t_1)$. Therefore,

$$\mathbb{P}(N(t_2) - N(t_1) = k) = e^{-\lambda(t_2 - t_1)} \frac{(\lambda(t_2 - t_1))^k}{k!}, \quad k = 0, 1, 2, \dots$$

Furthermore, if $0 < t_1 < t_2 < t_3 < t_4$, the random variables $N(t_2) - N(t_1)$ and $N(t_4) - N(t_3)$ are independent, and both are independent of $N(t_1)$. This characteristic is referred to as the *independent increments* property of the Poisson process. Additionally, if the intervals have equal duration ($t_2 - t_1 = t_4 - t_3 = \tau$), then $N(t_4) - N(t_3)$, $N(t_2) - N(t_1)$, and $N(\tau)$ are identically distributed random variables. This property characterises $N(t)$ as a stochastic process possessing *independent increments*.

1.2 First-order probability function

Let us now examine an alternative derivation of the first-order probability function of a Poisson process, based on three simple hypotheses regarding the intrinsic properties of the process. As previously defined, let $N(t)$ represent the number of transitions (or users arriving at the system) observed up to time t , where $N(0) = 0$. We will again focus on a subinterval of time with a duration of Δt . For clarity, we will also refer to this interval as Δt .

The hypotheses are the following (notice that (1) and (2) coincide, respectively, with (i) and (ii) given above).

- (1) $\mathbb{P}(\text{exactly 1 transition occurring within } \Delta t) = \lambda \Delta t + o(\Delta t)$.

In essence, the probability that one transition arrives in a subinterval Δt is proportional to Δt , with the constant of proportionality represented as λ . Notice that:

$$\lambda = \lim_{\Delta t \rightarrow 0} \frac{\mathbb{P}(\text{1 transition occurs in } \Delta t)}{\Delta t}.$$

- (2) $\mathbb{P}(\text{more than 1 transition occurring within } \Delta t) = o(\Delta t)$.

The probability that two or more transitions of the process arrive in the same time interval Δt is negligible compared to the probability of exactly one transition occurring.

- (3) *Independent increments*: If two time intervals do not overlap, the random variables representing the number of transitions appearing within these intervals are independent.

The hypotheses (1) and (2) indicate that the only two significant probabilities regarding the number of transitions observed in the interval Δt are:

$$\mathbb{P}(\text{1 transition in } \Delta t) \approx \lambda \Delta t, \quad \mathbb{P}(\text{no transitions in } \Delta t) \approx 1 - \lambda \Delta t.$$

First, we calculate the probability $\mathbb{P}(N(t) = 0)$ that, by time t , no transitions of the process have occurred. We have:

$$\begin{aligned} \mathbb{P}(N(t + \Delta t) = 0) &= \mathbb{P}(N(t) = 0, \text{ no transitions in } (t, t + \Delta t]) \\ &= \mathbb{P}(N(t) = 0) \mathbb{P}(\text{no transitions in } (t, t + \Delta t]) \\ &= \mathbb{P}(N(t) = 0)(1 - \lambda \Delta t + o(\Delta t)). \end{aligned}$$

In the above formulation we used that the events $N(t) = 0$ and “no transitions have occurred in $(t, t + \Delta t]$ ” are independent, because they involve the disjoint intervals $(0, t]$ and $(t, t + \Delta t]$ (see hypothesis (3)). Letting $g_0(t)$ represent $\mathbb{P}(N(t) = 0)$, we can rewrite the equation accordingly:

$$g_0(t + \Delta t) = g_0(t)(1 - \lambda \Delta t + o(\Delta t)).$$

Then,

$$\frac{g_0(t + \Delta t) - g_0(t)}{\Delta t} = g_0(t) \left(-\lambda + \frac{o(\Delta t)}{\Delta t} \right).$$

As we let $\Delta t \rightarrow 0$, and considering that $\lim_{\Delta t \rightarrow 0} o(\Delta t)/\Delta t = 0$, we derive the following first-order differential equation for the function $g_0(t)$ (which is linear, with constant coefficients, and homogeneous):

$$g_0'(t) + \lambda g_0(t) = 0, \quad t > 0.$$

Moreover, the initial condition for $g_0(t)$ is given by $g_0(0) = \mathbb{P}(N(0) = 0) = 1$, because we are assuming that $N(0) = 0$. The solution to this initial value problem is the function

$$g_0(t) = \mathbb{P}(N(t) = 0) = e^{-\lambda t}.$$

Hence, the probability of no transition having arrived by time t decreases exponentially as time progresses.

To find the expression for

$$g_k(t) = \mathbb{P}(N(t) = k), \quad k = 1, 2, \dots$$

we proceed in a similar way.

$$\begin{aligned} \mathbb{P}(N(t + \Delta t) = k) &= \mathbb{P}(N(t) = k, \text{ no transitions in } (t, t + \Delta t]) \\ &\quad + \mathbb{P}(N(t) = k - 1, \text{ exactly 1 transition in } (t, t + \Delta t]) \\ &\quad + \sum_{i=2}^k \mathbb{P}(N(t) = k - i, i \text{ transitions in } (t, t + \Delta t]). \end{aligned}$$

Taking into account the hypotheses (1), (2), and (3), we have:

$$g_k(t + \Delta t) = g_k(t)(1 - \lambda\Delta t + o(\Delta t)) + g_{k-1}(t)(\lambda\Delta t + o(\Delta t)) + o(\Delta t).$$

Dividing by Δt and letting $\Delta t \rightarrow 0$, we get a recurrent system of differential equations (of first order, linear, and with constant coefficients):

$$g'_k(t) + \lambda g_k(t) = \lambda g_{k-1}(t), \quad k = 1, 2, \dots, \quad (2)$$

where, now, the initial conditions to be considered are:

$$g_k(0) = \mathbb{P}(N(0) = k) = 0, \quad k = 1, 2, \dots. \quad (3)$$

This system can be easily solved using the Laplace transform method. With this aim, let $G_k(s)$ be the Laplace transform of the function $g_k(t)$. Transforming (2) and taking into account the initial conditions (3), we get:

$$(s + \lambda)G_k(s) = \lambda G_{k-1}(s).$$

Hence,

$$G_k(s) = \frac{\lambda}{s + \lambda} G_{k-1}(s) = \dots = \left(\frac{\lambda}{s + \lambda} \right)^k G_0(s) = \frac{\lambda^k}{(s + \lambda)^{k+1}},$$

where we have applied that $G_0(s) = 1/(s + \lambda)$ is the Laplace transform of the function $g_0(t) = e^{-\lambda t}$. Taking the inverse transformation, we finally have

$$g_k(t) = \mathbb{P}(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad (4)$$

as required.

Alternatively, the above system can be solved by assuming that the expression for $g_k(t)$ is given by (4) and, then, using induction from $g_0(t) = e^{-\lambda t}$.

1.3 The Markov property and the whole statistics of the process

In this subsection we will prove that a Poisson process $N(t)$ is a continuous-time Markov stochastic process, and as a consequence of this fact and the other properties exhibited before, the first-order probability function fully characterises the statistics of $N(t)$.

Theorem 1. *Let $N(t)$ be a Poisson process. For all $k \geq 2$ and all $0 < t_1 < t_2 < \dots < t_k$, we have*

$$\mathbb{P}(N(t_k) = n_k \mid N(t_{k-1}) = n_{k-1}, \dots, N(t_1) = n_1) = \mathbb{P}(N(t_k) = n_k \mid N(t_{k-1}) = n_{k-1}). \quad (5)$$

Hence, the stochastic process $N(t)$ has the Markov property.

Proof. Intuitively, we will demonstrate that both sides of (5) represent the probability that the number of arrivals within the interval $(t_{k-1}, t_k]$ is $n_k - n_{k-1}$.

The result is clearly valid for $k = 2$, so we assume $k \geq 3$. Without loss of generality, we suppose $0 < t_1 < t_2 < \dots < t_k$ and $0 \leq n_1 \leq n_2 \leq \dots \leq n_k$. We will utilise the fact that the increments of $N(t)$ are independent.

In one hand,

$$\begin{aligned}
& \mathbb{P}(N(t_k) = n_k \mid N(t_{k-1}) = n_{k-1}, \dots, N(t_1) = n_1) \\
&= \frac{\mathbb{P}(N(t_1) = n_1, \dots, N(t_{k-1}) = n_{k-1}, N(t_k) = n_k)}{\mathbb{P}(N(t_1) = n_1, \dots, N(t_{k-1}) = n_{k-1})} \\
&= \frac{\mathbb{P}(N(t_1) = n_1, N(t_2) - N(t_1) = n_2 - n_1, \dots, N(t_k) - N(t_{k-1}) = n_k - n_{k-1})}{\mathbb{P}(N(t_1) = n_1, N(t_2) - N(t_1) = n_2 - n_1, \dots, N(t_{k-1}) - N(t_{k-2}) = n_{k-1} - n_{k-2})} \\
&= \frac{\mathbb{P}(N(t_1) = n_1) \mathbb{P}(N(t_2) - N(t_1) = n_2 - n_1) \cdots \mathbb{P}(N(t_k) - N(t_{k-1}) = n_k - n_{k-1})}{\mathbb{P}(N(t_1) = n_1) \mathbb{P}(N(t_2) - N(t_1) = n_2 - n_1) \cdots \mathbb{P}(N(t_{k-1}) - N(t_{k-2}) = n_{k-1} - n_{k-2})} \\
&= \mathbb{P}(N(t_k) - N(t_{k-1}) = n_k - n_{k-1}).
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\mathbb{P}(N(t_k) = n_k \mid N(t_{k-1}) = n_{k-1}) &= \frac{\mathbb{P}(N(t_{k-1}) = n_{k-1}, N(t_k) = n_k)}{\mathbb{P}(N(t_{k-1}) = n_{k-1})} \\
&= \frac{\mathbb{P}(N(t_{k-1}) = n_{k-1}) \mathbb{P}(N(t_k) - N(t_{k-1}) = n_k - n_{k-1})}{\mathbb{P}(N(t_{k-1}) = n_{k-1})} \\
&= \mathbb{P}(N(t_k) - N(t_{k-1}) = n_k - n_{k-1}).
\end{aligned}$$

Thus we conclude that equality (5) holds for all $k \geq 2$ and all $0 < t_1 < t_2 < \dots < t_k$. $\square$

To provide a complete statistical characterization of a Poisson process $N(t)$ we must know all the joint probability functions $\mathbb{P}(N(t_1) = n_1, N(t_2) = n_2, \dots, N(t_k) = n_k)$.

Theorem 2. *Let $N(t)$ be a Poisson process. For all $k \geq 2$ and all $0 < t_1 < t_2 < \dots < t_k$, we have*

$$\begin{aligned}
& \mathbb{P}(N(t_1) = n_1, N(t_2) = n_2, \dots, N(t_k) = n_k) \\
&= \mathbb{P}(N(t_1) = n_1) \mathbb{P}(N(t_2) - N(t_1) = n_2 - n_1) \cdots \mathbb{P}(N(t_k) - N(t_{k-1}) = n_k - n_{k-1}).
\end{aligned}$$

Therefore, the first-order probability function determines the whole statistics of $N(t)$.

Proof. Firstly, notice that, as shown in the previous proof and by the stationary increments property, if $t_r < t_s$, then

$$\mathbb{P}(N(t_s) = n_s \mid N(t_r) = n_r) = \mathbb{P}(N(t_r) - N(t_s) = n_s - n_r) = \mathbb{P}(N(t_s - t_r) = n_s - n_r). \quad (6)$$

Then, by the Markov property and (6), we have:

$$\begin{aligned}
& \mathbb{P}(N(t_1) = n_1, \dots, N(t_k) = n_k) \\
&= \mathbb{P}(N(t_1) = n_1, \dots, N(t_{k-1}) = n_{k-1}) \mathbb{P}(N(t_k) = n_k \mid N(t_{k-1}) = n_{k-1}, \dots, N(t_1) = n_1) \\
&= \mathbb{P}(N(t_1) = n_1, \dots, N(t_{k-1}) = n_{k-1}) \mathbb{P}(N(t_k) = n_k \mid N(t_{k-1}) = n_{k-1}) \\
&= \mathbb{P}(N(t_1) = n_1, \dots, N(t_{k-2}) = n_{k-2}) \prod_{i=k-2}^{k-1} \mathbb{P}(N(t_{i+1}) = n_{i+1} \mid N(t_i) = n_i) \\
&= \dots \dots \\
&= \prod_{i=0}^{k-1} \mathbb{P}(N(t_{i+1}) = n_{i+1} \mid N(t_i) = n_i) = \prod_{i=0}^{k-1} \mathbb{P}(N(t_{i+1} - t_i) = n_{i+1} - n_i),
\end{aligned}$$

where, in the last line, $t_0 = 0$, $n_0 = 0$, and $\mathbb{P}(N(t_1) = n_1 \mid N(t_0) = 0) = \mathbb{P}(N(t_1) = n_1)$. $\square$

1.4 Expected value and autocorrelation function

As we know, for each fixed time $t \in (0, \infty)$, the random variable $N(t)$ derived from a Poisson process with intensity λ is Poisson-distributed with parameter λt . Therefore,

Theorem 3. *The mean value function of a Poisson process $N(t)$ with intensity λ is*

$$m_N(t) = \mathbb{E}[N(t)] = \lambda t, \quad t > 0.$$

Thus, the expected number of transitions arriving until the instant t increases linearly with t , being λ the proportionality factor. This result provides an alternative interpretation of this parameter:

$$\lambda = \frac{m_N(t)}{t}.$$

The value of λ represents the average number of transitions of the process per unit of time. In other words, λ indicates the transition rate. Notice that the Poisson process is not stationary, not even weakly (or in wide-sense) stationary.

To determine the autocorrelation function $R_N(t_1, t_2)$, let us assume first that $t_1 < t_2$. Then,

$$\begin{aligned} R_N(t_1, t_2) &= \mathbb{E}[N(t_1)N(t_2)] = \mathbb{E}[N(t_1)(N(t_1) + (N(t_2) - N(t_1)))] \\ &= \mathbb{E}[N^2(t_1)] + \mathbb{E}[N(t_1)(N(t_2) - N(t_1))] \\ &= (\lambda t_1)^2 + \lambda t_1 + \mathbb{E}[N(t_1)] \mathbb{E}[N(t_2) - N(t_1)] \\ &= (\lambda t_1)^2 + \lambda t_1 + \lambda t_1 \lambda (t_2 - t_1) = \lambda t_1 + \lambda^2 t_1 t_2. \end{aligned} \quad (7)$$

In the above computation, we used that the random variables $N(t_1)$ and $N(t_2) - N(t_1)$ are independent because the involved intervals, $(0, t_1]$ and $(t_1, t_2]$, do not overlap. We have also used that

$$\mathbb{E}[N^2(t_1)] = (\mathbb{E}[N(t_1)])^2 + \text{Var}[N(t_1)] = (\lambda t_1)^2 + \lambda t_1, \quad (8)$$

because $N(t_1) \sim \text{Po}(\lambda t_1)$ and so $\text{Var}[N(t_1)] = \lambda t_1$. In general, since $R_N(t_1, t_2) = R_N(t_2, t_1)$, and since (7) becomes (8) when $t_1 = t_2$, we have the following result:

Theorem 4. *The autocorrelation function of a Poisson process with intensity λ is:*

$$R_N(t_1, t_2) = \lambda \min(t_1, t_2) + \lambda^2 t_1 t_2, \quad t_1, t_2 > 0.$$

Consequently, the autocovariance function is given by:

$$C_N(t_1, t_2) = R_N(t_1, t_2) - m_N(t_1)m_N(t_2) = \lambda \min(t_1, t_2), \quad t_1, t_2 > 0.$$

1.5 Statistics of the inter-transition times

In this section, we consider some interesting properties regarding the random instants of time at which the transitions of the Poisson process occur.

Theorem 5. *If T is the random time at which the first transition of a Poisson process with intensity λ occurs, then $T \sim \text{Exp}(\lambda)$.*

Proof. Let $F_T(t) = \mathbb{P}(T \leq t)$ denote the cumulative distribution function of the random variable T . If $t < 0$, then $F_T(t) = \mathbb{P}(\emptyset) = 0$, whereas if $t > 0$:

$$\begin{aligned} F_T(t) &= \mathbb{P}(\text{at least 1 transition in } (0, t]) \\ &= 1 - \mathbb{P}(\text{no transitions in } (0, t]) = 1 - e^{-\lambda t} \frac{(\lambda t)^0}{0!} = 1 - e^{-\lambda t}. \end{aligned}$$

Then, the probability density function of T turns out to be $f_T(t) = F'_T(t) = \lambda e^{-\lambda t}$ for $t > 0$. $\square$

The above result also holds if T is the elapsed time between two consecutive transitions of the process.

Theorem 6. *The time interval between two consecutive transitions of a Poisson process with intensity λ follows an exponential distribution with parameter λ .*

Proof. Let U be the instant of the transition from which we begin to count the time T . As before, if $U = u$ and $t > 0$:

$$\begin{aligned} F_T(t | U = u) &= \mathbb{P}(T \leq t | U = u) = \mathbb{P}(\text{at least 1 transition in } (u, u + t]) \\ &= 1 - \mathbb{P}(\text{no transitions in } (u, u + t]) = 1 - e^{-\lambda t} \frac{(\lambda t)^0}{0!} = 1 - e^{-\lambda t}. \end{aligned}$$

Therefore, if $t > 0$,

$$F_T(t) = \int_0^\infty F_T(t | U = u) f_U(u) du = (1 - e^{-\lambda t}) \int_0^\infty f_U(u) du = 1 - e^{-\lambda t}.$$

□

More generally, we have the following result.

Theorem 7. *Let T_n represent the random time at which the n -th transition of a Poisson process with intensity λ occurs. More generally, T_n can be defined as the elapsed time between $n + 1$ consecutive transitions. The probability density function of the variable T_n is given by:*

$$f_{T_n}(t) = \begin{cases} 0, & t < 0, \\ \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}, & t > 0. \end{cases} \quad (9)$$

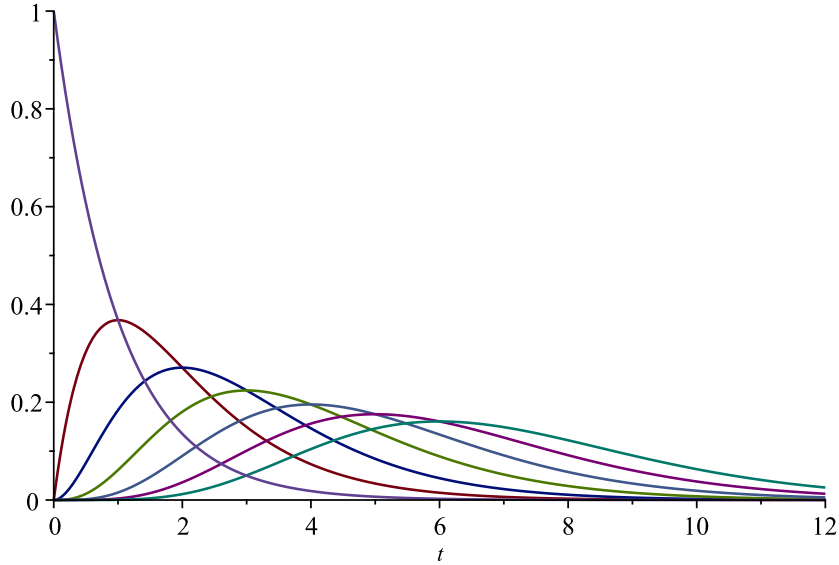


Figure 2: The density functions $f_{T_n}(t)$ for $n = 1, 2, \dots, 7$

Proof. Let T_n , for $n \geq 1$, be the random variable representing the time at which the n -th transition occurs. For any $t > 0$, we have:

$$\begin{aligned} F_{T_n}(t) &= \mathbb{P}(T_n \leq t) = \mathbb{P}(\text{at least } n \text{ transitions in } (0, t]) \\ &= 1 - \mathbb{P}(\text{less than } n \text{ transitions in } (u, u + t]) = 1 - \sum_{k=0}^{n-1} e^{-\lambda t} \frac{(\lambda t)^k}{k!}. \end{aligned} \quad (10)$$

If we consider T_n as the time elapsed between $n + 1$ consecutive transitions, we can easily see that (10) also holds. Specifically, if U is the random variable representing the time of the transition from which we start counting T_n , then

$$\begin{aligned} F_{T_n}(t | U = u) &= \mathbb{P}(T_n \leq t | U = u) \\ &= 1 - \mathbb{P}(\text{less than } n \text{ transitions in } (u, u + t]) = 1 - \sum_{k=0}^{n-1} e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \end{aligned}$$

and since this conditional distribution does not depend on u , we conclude that it is also equal to $F_{T_n}(t)$.

By differentiating, we deduce that the probability density function of T_n is given by the expression (9). (See Figure 2.) $\square$

The function described in (9) is the well-known **Gamma**(n, λ) density, and corresponds to the sum of n independent exponential random variables, each with parameter λ . This fact aligns with the statement given in Theorem 6.

The previous theorems demonstrate that if $\{T_n : n \geq 1\}$ is the sequence of random times at which transitions occur in a Poisson process with intensity λ , then the inter-transition times, specifically $X_1 = T_1$ and $X_n = T_n - T_{n-1}$ for $n \geq 2$, are random variables that follow an $\text{Exp}(\lambda)$ distribution. Additionally, the properties of the Poisson process imply that these variables are independent. To clarify this last point, let us see that the joint probability density of $X_1, X_2, \dots, X_n$ is given by $f_{X_1}(x_1)f_{X_2}(x_2) \cdots f_{X_n}(x_n)$. To this end, let us consider, for $0 < t_1 < t_2 < \cdots < t_n$, the following “incremental” probability:

$$\begin{aligned} f_T(t_1, t_2, \dots, t_n) \Delta t_1 \Delta t_2 \cdots \Delta t_n \\ \approx \mathbb{P}(t_1 \leq T_1 \leq t_1 + \Delta t_1, t_2 \leq T_2 \leq t_2 + \Delta t_2, \dots, t_n \leq T_n \leq t_n + \Delta t_n). \end{aligned} \quad (11)$$

If we ignore terms of order $o(\Delta t)$, the joint probability (11) is given by

$$e^{-\lambda t_1} \lambda \Delta t_1 \cdot e^{-\lambda(t_2 - t_1)} \lambda \Delta t_2 \cdots e^{-\lambda(t_n - t_{n-1})} \lambda \Delta t_n = \lambda^n e^{-\lambda t_n} \Delta t_1 \Delta t_2 \cdots \Delta t_n.$$

(This expression is valid because there is exactly one transition in each interval $(t_i, t_i + \Delta t_i]$, and no transitions occur in the intervals $(0, t_1]$ and $(t_{i-1}, t_i]$.) We conclude that the joint density of the variables $T_1, T_2, \dots, T_n$ is:

$$f_T(t_1, t_2, \dots, t_n) = \lambda^n e^{-\lambda t_n}, \quad 0 < t_1 < t_2 < \cdots < t_n.$$

Finally, since $X_1 = T_1$, $X_2 = T_2 - T_1$, $\dots$, and $X_n = T_n - T_{n-1}$, a straightforward change of variables shows that the joint probability density of the variables X_i is given for $x_1, x_2, \dots, x_n > 0$ by:

$$\begin{aligned} f_X(x_1, x_2, \dots, x_n) &= \lambda^n e^{-\lambda(x_1 + x_2 + \cdots + x_n)} \\ &= \lambda e^{-\lambda x_1} \lambda e^{-\lambda x_2} \cdots \lambda e^{-\lambda x_n} = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n), \end{aligned}$$

as we intended to demonstrate.

Conversely, we have the following result.

Theorem 8. Let $X_1, X_2, \dots, X_n, \dots$ be independent $\text{Exp}(\lambda)$ -distributed random variables. If $T_n = X_1 + \dots + X_n$ for $n \geq 1$, then, the stochastic process

$$N(t) = \max\{n: T_n \leq t\}, \quad t > 0,$$

which counts the number of random times T_n occurring in the interval $(0, t]$, is a Poisson process with intensity λ .

Proof. For any $k \geq 1$, consider the probability

$$\begin{aligned} \mathbb{P}(N(t) = k) &= \mathbb{P}(\text{exactly } k \text{ random times in } (0, t]) \\ &= \mathbb{P}(T_k \leq t, T_{k+1} > t) = \mathbb{P}(T_k \leq t, T_k + X_{k+1} > t) = \mathbb{P}((T_k, X_{k+1}) \in \mathcal{D}), \end{aligned}$$

where $\mathcal{D}$ is the region $\mathcal{D} = \{(\tau, \sigma) \in \mathbb{R}^2: 0 < \sigma < t, \tau + \sigma > t\}$ (see Figure 3).

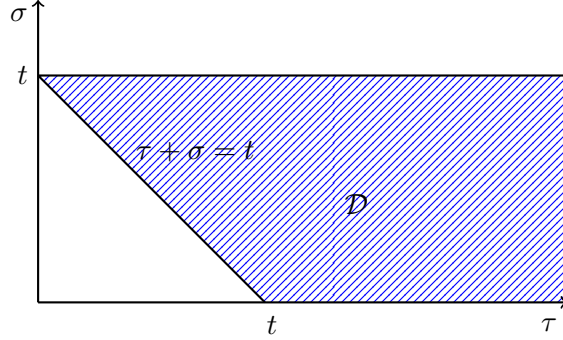


Figure 3: The integrating region $\mathcal{D}$.

Since T_k and X_{k+1} are independent (T_k depends only on $X_1, X_2, \dots, X_k$), we get:

$$\begin{aligned} \mathbb{P}(N(t) = k) &= \mathbb{P}((T_k, X_{k+1}) \in \mathcal{D}) = \iint_{\mathcal{D}} f_{T_k}(\sigma) f_{X_{k+1}}(\tau) d\sigma d\tau \\ &= \int_0^t \lambda e^{-\lambda\sigma} \frac{(\lambda\sigma)^{k-1}}{(k-1)!} \left(\int_{t-\sigma}^{\infty} \lambda e^{-\lambda\tau} d\tau \right) d\sigma \\ &= \int_0^t \lambda e^{-\lambda\sigma} \frac{(\lambda\sigma)^{k-1}}{(k-1)!} e^{-\lambda(t-\sigma)} d\sigma = e^{-\lambda t} \int_0^t \lambda \frac{(\lambda\sigma)^{k-1}}{(k-1)!} d\sigma \\ &= e^{-\lambda t} \frac{(\lambda t)^k}{k!}. \end{aligned}$$

Additionally, $\mathbb{P}(N(t) = 0) = \mathbb{P}(\text{no random times in } (0, t]) = \mathbb{P}(T_1 > t) = e^{-\lambda t}$. Consequently, $N(t)$ is a Poisson process with intensity λ . $\square$

Another interesting aspect of the transitions in a Poisson process is the following: Assume there has been exactly one transition in the interval $(0, t]$. In this case, the random variable T , which represents the time at which the transition occurs, follows a uniform distribution between 0 and t . Specifically, if $\tau \in (0, t]$, then

$$\begin{aligned} \mathbb{P}(T \leq \tau | N(t) = 1) &= \frac{\mathbb{P}(T \leq \tau, N(t) = 1)}{\mathbb{P}(N(t) = 1)} = \frac{\mathbb{P}(N(\tau) = 1, N(t) - N(\tau) = 0)}{\mathbb{P}(N(t) = 1)} \\ &= \frac{\mathbb{P}(N(\tau) = 1) \mathbb{P}(N(t) - N(\tau) = 0)}{\mathbb{P}(N(t) = 1)} = \frac{\mathbb{P}(N(\tau) = 1) \mathbb{P}(N(t - \tau) = 0)}{\mathbb{P}(N(t) = 1)} \\ &= \frac{\lambda \tau e^{-\lambda\tau} \cdot e^{-\lambda(t-\tau)}}{\lambda t e^{-\lambda t}} = \frac{\tau}{t}. \end{aligned}$$

See Example 1 for a generalization of this property.

1.6 Marked transitions.

We know that if $\{X_n : n \geq 1\}$ is a sequence of independent Bernoulli random variables with parameter p (where $0 < p < 1$), and N is a random variable that follows a Poisson distribution with parameter μ and is independent of the X_n , then the sum $X_1 + X_2 + \cdots + X_N$ (where N indicates a random number of terms) is also a Poisson-distributed random variable, and its parameter is μp . Consequently, if $N(t)$ is a Poisson process, we have the following result.

Theorem 9. *If the transitions of a Poisson process $N(t)$ with intensity λ are independently marked with probability p , $0 < p < 1$, then the number $N'(t)$ of marked transitions in $(0, t]$ defines a Poisson process with intensity $\lambda' = \lambda p$.*

Notice that this result is consistent with the hypotheses regarding the Poisson process. To illustrate, let us consider a small time interval Δt . There are two significant possibilities: either we observe exactly one marked transition occurring within Δt , and this event happens with a probability of $\lambda p \Delta t + o(\Delta t)$, or we do not observe any marked transitions within Δt , which occurs with a probability of $1 - \lambda p \Delta t + o(\Delta t)$. (The likelihood of observing more than one marked transition in Δt is of the order $o(\Delta t)$, and so it is negligible.) Furthermore, if A and B represent non-overlapping time intervals, then the number of marked transitions observed in interval A is independent of the number of marked transitions in interval B .

1.7 Compound Poisson processes.

If X and Y are independent Poisson random variables with parameters μ_X and μ_Y , respectively, then the sum $X + Y$ also follows a Poisson distribution with parameter $\mu_X + \mu_Y$. A similar result applies to Poisson processes. Two Poisson processes $N_1(t)$ and $N_2(t)$ are defined as independent if the random variables $N_1(t)$ and $N_2(t')$ are independent for all $t, t' \geq 0$.

Theorem 10. *If $N_1(t)$ and $N_2(t)$ are independent Poisson processes with intensities λ_1 and λ_2 , respectively, then the compound process $N_1(t) + N_2(t)$ is a Poisson process with intensity $\lambda_1 + \lambda_2$.*

1.8 Some examples.

In this last subsection, let us consider several exercises regarding the Poisson process.

Example 1. *A computer's processing unit requests memory services at random times, which can be modeled as a Poisson process. The number of requests, denoted as $N(t)$, occurring in the interval $(0, t]$ follows a Poisson distribution with an average request rate of λ services per unit time.*

Given that $n \geq 1$ services have been requested in the interval $(0, t]$, the objective is to determine the probability that exactly k services were requested in the subinterval $(0, \tau]$, where $0 < \tau < t$.

Solution. We have to determine the probability $\mathbb{P}(N(\tau) = k \mid N(t) = n)$. If $k > n$, the probability

is zero by definition. Therefore, it is sufficient to consider the case where $0 \leq k \leq n$, in which case:

$$\begin{aligned}
\mathbb{P}(N(\tau) = k \mid N(t) = n) &= \frac{\mathbb{P}(N(\tau) = k, N(t) = n)}{\mathbb{P}(N(t) = n)} \\
&= \frac{\mathbb{P}(N(\tau) = k, N(t) - N(\tau) = n - k)}{\mathbb{P}(N(t) = n)} \\
&= \frac{\mathbb{P}(N(\tau) = k) \mathbb{P}(N(t) - N(\tau) = n - k)}{\mathbb{P}(N(t) = n)} \\
&= \frac{\mathbb{P}(N(\tau) = k) \mathbb{P}(N(t - \tau) = n - k)}{\mathbb{P}(N(t) = n)} \\
&= \frac{(e^{-\lambda\tau}(\lambda\tau)^k/k!) (e^{-\lambda(t-\tau)}(\lambda(t-\tau))^{n-k}/(n-k)!)}{e^{-\lambda t}(\lambda t)^n/n!} \\
&= \binom{n}{k} \left(\frac{\tau}{t}\right)^k \left(1 - \frac{\tau}{t}\right)^{n-k}.
\end{aligned}$$

Thus, the resulting probability follows a binomial distribution with parameters n and $p = \tau/t$.

Example 2. *Passengers arrive at a bus stop according to a Poisson process with rate $\lambda = 5$ passengers per minute. A bus arrives at the stop every 4 minutes.*

- (a) *Suppose a passenger arrives at the bus stop and observes that 4 people are already waiting. Determine the expected waiting time for this passenger.*
- (b) *When the bus arrives, there are 20 people at the stop. Demonstrate that the random variable representing the number of travellers who have been waiting for three or more minutes follows a binomial distribution. Additionally, calculate the expected number of passengers who have been waiting for at least three minutes.*

Solution. Let $N(t)$ represent the number of passengers arriving in the time interval $(0, t]$, where $t = 0$ marks the moment when the last bus left the stop.

- (a) When a passenger arrives at the bus stop and observes 4 individuals already waiting, the arrival time, denoted as X_5 , represents the occurrence of the fifth transition in the process $N(t)$. Given that buses arrive every 4 minutes, the waiting time for this passenger is expressed as $T = 4 - X_5$. The variable X_5 is the sum of five independent random variables, each distributed as $\text{Exp}(5)$. Thus, the expected value of X_5 is $\mathbb{E}[X_5] = 5 \cdot (1/5) = 1$ minute. Accordingly, the expected waiting time T for the passenger is $\mathbb{E}[T] = 4 - \mathbb{E}[X_5] = 3$ minutes.
- (b) The condition that there are k people at the bus stop who have been waiting for at least 3 minutes is equivalent to the event $N(1) = k$. The objective is to determine the probability of this event, given that 20 passengers are present when the bus arrives at $t = 4$. The argument

proceeds analogously to the previous example:

$$\begin{aligned}
\mathbb{P}(N(1) = k \mid N(4) = 20) &= \frac{\mathbb{P}(N(1) = k, N(4) = 20)}{\mathbb{P}(N(4) = 20)} \\
&= \frac{\mathbb{P}(N(1) = k) \mathbb{P}(N(4) - N(1) = 20 - k)}{\mathbb{P}(N(4) = 20)} \\
&= \frac{\mathbb{P}(N(1) = k) \mathbb{P}(N(3) = 20 - k)}{\mathbb{P}(N(4) = 20)} \\
&= \frac{(e^{-\lambda} \lambda^k / k!) (e^{-3\lambda} (3\lambda)^{20-k} / (20-k)!)}{e^{-4\lambda} (4\lambda)^{20} / 20!} \\
&= \binom{20}{k} \left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{20-k}, \quad 0 \leq k \leq 20.
\end{aligned}$$

Therefore, the number of passengers who have waited at least three minutes, given that 20 people are present when the bus arrives, follows a $\text{Bin}(20, 1/4)$ distribution. The expected value of this quantity is $20 \cdot (1/4) = 5$ people.

Example 3. Consider a Poisson process $N(t)$ with intensity λ . Suppose each transition is independently marked with probability p , where $0 < p < 1$. Determine the probability that no marked transitions occur in the interval $(0, t]$. Additionally, compute the expected time until the first marked transition occurs.

Solution. The marked process $N'(t)$ is a Poisson process with intensity λp . Therefore, the probability that there are no marked transitions in $(0, t]$ is $\mathbb{P}(N'(t) = 0) = e^{-\lambda p t}$. This result can also be derived directly as follows:

$$\begin{aligned}
\mathbb{P}(N'(t) = 0) &= \sum_{n=0}^{\infty} \mathbb{P}(N'(t) = 0 \mid N(t) = n) \mathbb{P}(N(t) = n) \\
&= \sum_{n=0}^{\infty} (1-p)^n e^{-\lambda t} \frac{(\lambda t)^n}{n!} = e^{-\lambda t} \sum_{n=0}^{\infty} \frac{((1-p)\lambda t)^n}{n!} \\
&= e^{-\lambda t} \cdot e^{(1-p)\lambda t} = e^{-\lambda p t}.
\end{aligned}$$

Since $N'(t)$ has intensity λp , the time T' until the first marked transition occurs is exponentially distributed with parameter λp . Therefore, $\mathbb{E}[T'] = 1/(\lambda p)$. The cumulative distribution function of the random variable T' is given by

$$F_{T'}(t) = P(T' \leq t) = 1 - \mathbb{P}(T' > t) = 1 - \mathbb{P}(N'(t) = 0) = 1 - e^{-\lambda p t}, \quad t > 0,$$

which implies that $T' \sim \text{Exp}(\lambda p)$.

Example 4. Consider an email account where the time between two consecutive received messages is modelled as an exponential random variable with an expected value of $5/3$ minutes. The account is checked every five minutes.

- What is the expected number of incoming messages present in the account at each check?
- Is it more probable that the user finds more messages than the expected value, or fewer messages?
- If one third of incoming messages are classified as spam, what is the probability that more than one spam message is found during a single account check?

Solution. The arrival of incoming messages is modelled as a Poisson process $N(t)$ with intensity $\lambda = 3/5$ messages per minute.

- (a) The number $N(5)$ of messages received over a five-minute interval is a Poisson random variable with parameter $5\lambda = 3$. Therefore, the expected value is $\mathbb{E}[N(5)] = 3$.
- (b) The probability of receiving fewer messages than the expected value is given by

$$\mathbb{P}(N(5) < 3) = \sum_{k=0}^2 \mathbb{P}(N(5) = k) = e^{-3} \left(1 + 3 + \frac{9}{2} \right) \approx 0.423.$$

Conversely,

$$\mathbb{P}(N(5) > 3) = 1 - \mathbb{P}(N(5) \leq 3) = 1 - \sum_{k=0}^3 \mathbb{P}(N(5) = k) = 1 - e^{-3} \left(1 + 3 + \frac{9}{2} + \frac{27}{6} \right) \approx 0.353.$$

Thus, it is more probable to receive fewer messages than the expected value.

- (c) The number of spam messages, denoted as $N'(t)$, received in the interval $(0, t]$ also follows a Poisson process, with intensity $\lambda' = \lambda \cdot \frac{1}{3} = \frac{1}{5}$. This results from independently marking each transition of $N(t)$ with probability $p = \frac{1}{3}$. Therefore, the probability that more than one spam message is found when checking the account is

$$\mathbb{P}(N'(5) > 1) = 1 - \mathbb{P}(N'(5) \leq 1) = 1 - \mathbb{P}(N'(5) = 0) - \mathbb{P}(N'(5) = 1) = 1 - 2e^{-1} \approx 0.264.$$

Example 5. *Each customer's service time at a bank counter is independently distributed according to an exponential distribution with a mean of 2 minutes. Upon arrival at the counter, there are 3 customers present.*

- (a) *What is the probability that, after 6 minutes, service is being provided to you at the counter?*
- (b) *Given that the first customer remains at the counter after 3 minutes, what is the probability that, 6 minutes later, the customer immediately ahead in line is still waiting?*
- (c) *After observing that the first customer has been at the counter for 10 minutes, the bank opens a second counter to assist those waiting in line. What is the probability that you will be receiving service 6 minutes later? Is this probability greater or less than the probability calculated in part (a)?*

Solution. Let the origin of time, $t = 0$, correspond to the moment of arrival. Since the service time T is an exponential random variable, it possesses the memoryless property. Therefore, at $t = 0$, the situation is equivalent to a customer beginning service upon arrival. In this scenario, one person is initiating service, with two additional customers waiting in the queue.

Assuming a continuous arrival of customers and that the queue is never empty, the number $N(t)$ of customers served in the interval $(0, t]$ follows a Poisson process with intensity $\lambda = 1/\mathbb{E}[T] = 1/2$ customers per minute.

- (a) If after 6 minutes we are currently being served, then $N(6) = 3$. Since $N(6) \sim \text{Po}(6\lambda)$, the probability of this event is

$$\mathbb{P}(N(6) = 3) = e^{-3} \frac{3^3}{3!} \approx 0.224.$$

- (b) The exponential random variable is defined by its memoryless property. If the first customer remains at the counter after three minutes, the situation is statistically equivalent to the initial observation. Consequently, the probability that the customer in front remains after an additional six minutes is

$$\mathbb{P}(N(6) \leq 1) = \mathbb{P}(N(6) = 0) + \mathbb{P}(N(6) = 1) = e^{-3}(1 + 3) \approx 0.199.$$

- (c) The total number of customers served in the interval $(0, t]$ is now given by $N(t) = N_1(t) + N_2(t)$, where $N_i(t)$ denotes the number of customers served at counter i for $i = 1, 2$. Because both processes are independent and identically distributed, the aggregate process $N(t)$ forms a Poisson process with total intensity $\lambda = \lambda_1 + \lambda_2 = 1$.

Now the probability that 6 minutes later we will be receiving service is:

$$\mathbb{P}(N(6) = 3) = e^{-6} \frac{6^3}{3!} \approx 0.089.$$

It is important to note that this probability, which is significantly smaller than the value computed previously in (a), does not imply a longer waiting time. Rather, it suggests that it is more likely the customer will have already been served after six minutes.

Example 6. *Emergencies arrive at a hospital according to a Poisson process with a rate of 4 arrivals per minute.*

- (a) *Given that no emergencies have occurred at time $t = 0$, determine the probability that the first emergency occurs after $t = 1/2$ minute.*
- (b) *If two emergencies have already arrived by $t = 1$ minute, calculate the probability that the second emergency occurred more than $1/2$ minute prior to $t = 1$.*
- (c) *Assume that the service time for each emergency is exponentially distributed with a mean of $1/4$ minute. If the first emergency attendance begins at $t = 1$ and two emergencies have already arrived by this time, determine the expected number of additional emergencies that will arrive between $t = 1$ and the completion of service for the first two emergencies.*

Solution. Let $N(t)$ be the number of emergencies that have arrived up to instant t . Then $N(t)$ is a Poisson process with rate $\lambda = 4$.

- (a) Let T_1 denote the arrival time of the first emergency. Then,

$$\mathbb{P}(T_1 > 1/2) = \mathbb{P}(N(1/2) = 0) = e^{-\lambda/2} = e^{-2} \approx 0.135.$$

- (b) Let T_2 be the arrival time of the second emergency. Therefore,

$$\begin{aligned} \mathbb{P}(T_2 \leq 1/2 | N(1) = 2) &= \frac{\mathbb{P}(T_2 \leq 1/2, N(1) = 2)}{\mathbb{P}(N(1) = 2)} = \frac{\mathbb{P}(N(1/2) = 2, N(1) = 2)}{\mathbb{P}(N(1) = 2)} \\ &= \frac{\mathbb{P}(N(1/2) = 2, N(1) - N(1/2) = 0)}{\mathbb{P}(N(1) = 2)} = \frac{\mathbb{P}(N(1/2) = 2) \mathbb{P}(N(1) - N(1/2) = 0)}{\mathbb{P}(N(1) = 2)} \\ &= \frac{\mathbb{P}(N(1/2) = 2) \mathbb{P}(N(1/2) = 0)}{\mathbb{P}(N(1) = 2)} = \frac{(\lambda/2)^2 e^{-\lambda/2} / 2 \cdot e^{-\lambda/2}}{\lambda^2 e^{-\lambda/2}} = \frac{1}{4}, \end{aligned}$$

where we have used the independence and homogeneity regarding disjoint intervals of time.

- (c) Let S_1 and S_2 denote the attendance times for the first and second emergencies, respectively. The total attendance time is given by $S = S_1 + S_2$. Thus, $\mathbb{E}[S] = \mathbb{E}[S_1] + \mathbb{E}[S_2] = 1/2$ minute. The expected number of emergencies that arrive from $t = 1$ until the first two emergencies have been attended is given by $\mathbb{E}[N(S)]$. By conditioning on S and applying the law of total expectation, it follows that

$$\mathbb{E}[N(S)] = \mathbb{E}[\mathbb{E}[N(S) | S]],$$

where,

$$\mathbb{E}[N(S) | S = s] = \mathbb{E}[N(s)] = \lambda s.$$

Therefore, it can be concluded that

$$\mathbb{E}[N(S)] = \mathbb{E}[\lambda S] = \lambda \mathbb{E}[S] = 2.$$

Example 7. Consider a Poisson process $N(t)$ with intensity λ . Define the random variables $Y_1 = N(t_2) - N(t_1)$ and $Y_2 = N(t_3) - N(t_2)$, where $0 \leq t_1 < t_2 < t_3$. Determine $\mathbb{E}[Y_1 Y_2]$. Additionally, evaluate the result when $t_1 = 0$.

Solution. If $N(t)$ is a Poisson process with intensity λ , we know that for any $0 \leq a < b$, the random variable $N(b) - N(a)$ follows a Poisson distribution with parameter $\lambda(b - a)$. Consequently, we have $\mathbb{E}[N(b) - N(a)] = \lambda(b - a)$ and $\mathbb{E}[(N(b) - N(a))^2] = \lambda(b - a) + \lambda^2(b - a)^2$. Moreover, if $0 \leq a < b < c < d$, then the random variables $N(d) - N(c)$ and $N(b) - N(a)$ are independent.

We have

$$\begin{aligned} \mathbb{E}[Y_1 Y_2] &= \mathbb{E}[(N(t_2) - N(t_1))(N(t_3) - N(t_2))] \\ &= \mathbb{E}[(N(t_2) - N(t_1))(N(t_1) + (N(t_2) - N(t_1)) + (N(t_3) - N(t_2)))] \\ &= \mathbb{E}[(N(t_2) - N(t_1))N(t_1)] + \mathbb{E}[(N(t_2) - N(t_1))^2] + \mathbb{E}[(N(t_2) - N(t_1))(N(t_3) - N(t_2))] \\ &= \mathbb{E}[N(t_2) - N(t_1)]\mathbb{E}[N(t_1)] + \mathbb{E}[(N(t_2) - N(t_1))^2] + \mathbb{E}[N(t_2) - N(t_1)]\mathbb{E}[N(t_3) - N(t_2)] \\ &= \lambda(t_2 - t_1)\lambda t_1 + (\lambda(t_2 - t_1) + \lambda^2(t_2 - t_1)^2) + \lambda(t_2 - t_1)\lambda(t_3 - t_2) \\ &= \lambda(t_2 - t_1) + \lambda^2(t_1 t_2 - t_1^2 + t_1^2 - 2t_1 t_2 + t_2^2 + t_2 t_3 - t_1 t_3 - t_2^2 + t_1 t_2) \\ &= \lambda(t_2 - t_1) + \lambda^2(t_2 - t_1)t_3 = \lambda(t_2 - t_1)(1 + \lambda t_3). \end{aligned}$$

When $t_1 = 0$, we obtain the autocorrelation function $R_N(t_2, t_3)$.

Example 8. Let $N(t)$ be a Poisson process with intensity λ .

- (a) Determine the optimal homogeneous linear estimation of $N(2)$ based on $N(1)$ and $N(3)$, specifically, $\widehat{N(2)} = \alpha N(1) + \beta N(3)$. Provide an analysis of the resulting estimator.
- (b) Determine the optimal non-homogeneous linear estimation of $N(5)$ in terms of $N(4)$, that is, $\widehat{N(5)} = \alpha N(4) + \beta$. Discuss the obtained result.

Solution.

- (a) If $\widehat{N(2)} = \alpha N(1) + \beta N(3)$ represents the optimal homogeneous linear estimator of the random variable $N(2)$ in the quadratic mean, then the coefficients α and β are determined by the following orthogonality conditions:

$$\mathbb{E}[(N(2) - \widehat{N(2)})N(1)] = 0, \quad \mathbb{E}[(N(2) - \widehat{N(2)})N(3)] = 0.$$

The values of α and β obtained from these conditions guarantee minimisation of the mean squared error $\mathbb{E}[(N(2) - \widehat{N(2)})^2]$.

Hence, taking into account that $\mathbb{E}[N(t_1)N(t_2)] = R_N(t_1, t_2)$, we have to solve the linear system:

$$\begin{aligned}\alpha R_N(1, 1) + \beta R_N(1, 3) &= R_N(1, 2) \\ \alpha R_N(1, 3) + \beta R_N(3, 3) &= R_N(2, 3)\end{aligned}$$

If $t_1 < t_2$, the autocorrelation function of $N(t)$ is given by $R_N(t_1, t_2) = \lambda t_1 + \lambda^2 t_1 t_2$. Thus, after simplifying by λ ,

$$\begin{aligned}(1 + \lambda)\alpha + (1 + 3\lambda)\beta &= 1 + 2\lambda \\ (1 + 3\lambda)\alpha + (3 + 9\lambda)\beta &= 2 + 6\lambda\end{aligned}$$

It is easily checked that the solution is $\alpha = \beta = 1/2$. Therefore,

$$\widehat{N(2)} = \frac{N(1) + N(3)}{2}. \quad (12)$$

Since the expected number of transitions in an interval of length one is λ , we can estimate $N(2) \approx N(1) + \lambda$, and, similarly, $N(2) \approx N(3) - \lambda$. We deduce that

$$\lambda \approx \frac{N(3) - N(1)}{2}.$$

and so

$$N(2) \approx N(1) + \lambda \approx N(1) + \frac{N(3) - N(1)}{2} = \frac{N(1) + N(3)}{2},$$

in accordance with (12).

- (b) Let $\widehat{N(5)} = \alpha N(4) + \beta$. Now, the orthogonality conditions to find the values of α and β that minimize the the mean squared error $\mathbb{E} \left[\left(N(5) - \widehat{N(5)} \right)^2 \right]$ are:

$$\mathbb{E} \left[N(5) - \widehat{N(5)} \right] = 0, \quad \mathbb{E} \left[\left(N(5) - \widehat{N(5)} \right) N(4) \right] = 0.$$

We have

$$\begin{aligned}\mathbb{E} (N(5) - \alpha N(4) - \beta) &= 0 \\ \mathbb{E} ((N(5) - \alpha N(4) - \beta) N(4)) &= 0\end{aligned}$$

Therefore,

$$\begin{aligned}4\lambda\alpha + \beta &= 5\lambda \\ (4\lambda + 16\lambda^2)\alpha + 4\lambda\beta &= 4\lambda + 20\lambda^2\end{aligned}$$

The solution to this system is $\alpha = 1$ and $\beta = \lambda$. Hence,

$$\widehat{N(5)} = N(4) + \lambda.$$

Let $Y = N(5) - N(4)$ denote the number of transitions occurring in the interval $(4, 5]$. Given $N(4)$, $N(5)$ can be estimated by adding the expected value of Y to $N(4)$. Due to the properties of the Poisson process, $N(4)$ and Y are independent random variables. Consequently, the optimal estimate (in the quadratic mean) of Y given $N(4)$ is simply $\mathbb{E}(Y) = \lambda$. Thus, the estimated value of $N(5)$ is $\widehat{N(5)} = N(4) + \lambda$.